

them with offsetting stock positions, an approach they detailed in their book, *Beat the Market*.³

- He was the first to formulate an option-pricing model that was equivalent to the Black-Scholes model. Thorp had actually used an equivalent form of the formula to very profitably trade warrants and options for years before the publication of the Black-Scholes model.
- He was the founder of the first market neutral fund.
- He established the first successful quant hedge fund.
- He was the first to implement convertible arbitrage.
- He was the first to implement statistical arbitrage.
- He was likely the first person to uncover that Bernie Madoff was a fraud—he developed conclusive evidence of the fraud many years before Harry Markopolos did.⁴

Thorp, a PhD mathematician, and near PhD physicist, came to the markets via gambling, but not gambling in the conventional sense. Normally, casino games of chance have a negative edge for the player, and the longer one plays, the greater the chance of financial ruin. This type of gambling is the antithesis of what Thorp was interested in. Thorp was, in fact, extremely risk averse, a by-product of his having grown up during the Depression. Thorp's goal was to remove the gambling from gambling. He sought to devise strategies that would place the edge in casino games in his favor—a task that had been assumed to be impossible. Amazingly, he was successful in devising strategies to gain a significant edge in multiple casino games including roulette, blackjack, baccarat, and Wheel of Fortune. Ironically, devising the strategies to win at what were always assumed to be unbeatable games proved to be easier than the execution. Winning from casinos presents practical problems. Winning players get noticed, and casinos have low tolerance for players who win by any means other than pure chance.

Thorp thought that the markets might provide a better alternative to apply his research. The markets were after all the largest game, and no one could kick him out if he figured out a consistent way to win. So he turned his research focus on the stock market. This research led to his discovery that warrants (long-term options) were mispriced. In working on the problem of how to price warrants and options, Thorp was introduced to Sheen Kassouf,